ARCHITECTURAL SPECIFICATION

Amagi-FMEDA v1.0

Failure Modes, Effects, and Diagnostic Analysis

VERSION 1.0 — JUNE 2026

Quantitative safety integrity analysis for the AMAGI hardware TCB per IEC 61508-2. Nine subsystem components analyzed: λ_DU = 18.9 FIT (component-level), SFF = 93.3% (system-level with CRTM/EGW), PFDavg = 6.9×10^{-5} (1oo1, SIL 2) / 6.9×10^{-6} (1oo2, SIL 3). Software/AI classified as Certain Event (P≈1), excluded from Safety-Critical Path by INV-003 + INV-005 (defined in NIPU v1.1). Aligned with: Framework v2.0, NIPU v1.1, HARA v1.0, Traceability Matrix v1.0, Master Parameter Sheet v2.0, S-APL v2.0, Charter v1.2.

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DOI: 10.5281/zenodo.20753410

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1. Executive Summary

This Failure Modes, Effects, and Diagnostic Analysis (FMEDA) provides the component-level failure analysis for the AMAGI Architectural Class hardware, conducted in accordance with IEC 61508-2 §7.4.6 and §7.4.9. The FMEDA is a systematic, bottom-up methodology that identifies each hardware component within the Target of Evaluation (TOE), determines its potential failure modes, classifies each failure as safe or dangerous (detectable or undetectable), and computes the aggregate safety metrics required to demonstrate compliance with the target Safety Integrity Level (SIL). This document serves as the quantitative safety evidence supporting the AMAGI safety argument and is intended for submission to certification bodies evaluating the AMAGI architecture against IEC 61508 requirements.

The FMEDA quantifies the failure rates in FIT (Failures In Time, where 1 FIT = 1×10^{-9} failures per hour), the diagnostic coverage (DC) for each subsystem, and the Safe Failure Fraction (SFF) across the entire AMAGI hardware architecture. The reference implementation target is a Xilinx Artix-7 FPGA with 2,500–3,200 LUTs operating at 100 MHz (post-synthesis 142 MHz) and consuming approximately 1.3W. The analysis covers six subsystem components per the FMEDA breakdown aligned with the TCAD reference article: the SCC FSM (core), TTP delay chain, MEM/MAG (BRAM+OTP), Amagi-Q (PQC core), optocoupled I/O, and clock/power isolation.

The overall results demonstrate compliance with SIL 2 requirements under the 1oo1 voting architecture and SIL 3 under the 1oo2 voting architecture. The component-level SFF is 91.6% (system-level SFF = 93% with CRTM/EGW diagnostic uplift per ADR-006/007), which satisfies the ≥90% threshold for Route 2_H with Hardware Fault Tolerance (HFT) = 1. The average Probability of Dangerous Failure (PFD_{avg}) for the 1oo1 architecture is 6.9×10^{-5}, achieving SIL 2 per IEC 61508-3 Clause 7.3 (10^{-3} to 10^{-4}). The 1oo2 architecture achieves $PFD_{avg} = 6.9\times10^{-6}$, providing deep SIL 3 compliance with substantial margin. The component-level dangerous undetected failure rate λ_{du} = 18.9 FIT (system-level ~15.75 FIT after CRTM/EGW uplift) is dominated by the optocoupled I/O (8.4 FIT) and TTP delay chain (4.5 FIT), which together account for approximately 68% of the total component-level dangerous undetected failures.

2. Scope and Applicable Standards

This FMEDA is conducted in accordance with the following international standards, which provide the normative framework for functional safety assessment of electrical, electronic, and programmable electronic (E/E/PE) safety-related systems. The primary standard governing the FMEDA methodology and acceptance criteria is IEC 61508-2:2010, which defines the requirements for hardware safety integrity of safety-related systems. Specifically, IEC 61508-2 §7.4.6 mandates the identification of all relevant failure modes and the estimation of safe and dangerous failure rates, while §7.4.9 specifies the requirements for diagnostic coverage and the calculation of the Safe Failure Fraction.

The failure rate prediction methodology follows IEC 62380:2004 (Reliability data handbook — Universal model for prediction of reliability on electronic components), which provides component-level base failure rates adjusted for operating conditions including ambient temperature, thermal cycling, and mission profile. IEC 61508-2 Annex A provides the diagnostic coverage reference tables used to assign conservative DC values to each diagnostic mechanism based on the specific technique employed. These three standards together form the complete normative basis for the quantitative analysis presented in this document.

2.1 Applicable Standards

Standard	Year	Scope	Application in FMEDA
IEC 61508-2	2010	Functional safety of E/E/PE systems	FMEDA methodology, SFF/PFD calculation, SIL verification
IEC 62380	2004	Failure rate prediction for electronic components	Component base failure rates (FIT)
IEC 61508-2 Annex A	2010	Diagnostic coverage tables	DC assignment for each diagnostic technique

Table 1: Applicable standards

2.2 TOE Coverage

The FMEDA covers the complete AMAGI Target of Evaluation (TOE), which consists of the following six subsystem groupings per the TCAD reference article: (1) the SCC FSM (core), comprising the Safety Control Core Finite State Machine and its associated control logic; (2) the TTP delay chain, comprising the PVT-calibrated DDL and Actuation Time Lock that enforce the Tautological Truth Predicate; (3) MEM/MAG (BRAM+OTP), combining the Secure Hardware Memory Firewall with ECC, the Mutable Authenticated Graph OTP array, and the Code Root of Trust Mask ROM; (4) the Amagi-Q Post-Quantum Cryptography core; (5) the optocoupled I/O subsystem, comprising the Emergency Gate Watchdog AND-gate, the audit diode, and all galvanically-isolated input/output interfaces; and (6) the clock/power isolation subsystem, comprising the Analog Fallback Circuit, independent LDO, and clock distribution network. All six subsystem groupings are implemented within the Xilinx Artix-7 FPGA fabric, with CRTM realized as a mask-ROM region and MAG OTP realized as one-time programmable fuses or antifuses.

2.3 Exclusions

The following elements are excluded from the FMEDA scope, as permitted by IEC 61508-2 §7.4.2.1 and justified by the AMAGI safety argument. Cognitive clusters (the Normal World processing elements including GEN, COM, ANL, and EXE) are excluded because they reside outside the TOE boundary; the AMAGI invariants do not depend on their correct operation, and the EGW AND-gate ensures that cognitive cluster failure cannot cause dangerous actuation. The power supply is excluded per IEC 61508-2 §7.4.2.1, which allows the assumption of a reliable power supply when the supply is separately certified or when the safety function response to power loss is defined; the AFC provides the defined safe-state response to power loss. The external 2FA signal is assumed to be available as an independent safety input, per the EGW AND-gate design where Actuator_Power = SCC_Enable AND Ext_2FA_Enable.

3. FMEDA Methodology

The FMEDA process follows the systematic procedure defined in IEC 61508-2 §7.4.6, which requires a bottom-up analysis of each hardware component to identify all relevant failure modes, classify them as safe or dangerous, and determine the extent to which dangerous failures can be detected by automatic diagnostic measures. The methodology consists of six sequential steps, each producing a specific output that feeds into the subsequent step. The results of this process form the quantitative basis for demonstrating that the AMAGI hardware architecture achieves the required Safety Integrity Level.

Step 1: Component Identification. Each hardware component within the AMAGI TOE is identified and catalogued, including its function, technology type (combinatorial logic, sequential logic, mixed-signal, analog, OTP, ROM, SRAM), and implementation technology (FPGA LUT, mask-ROM, OTP, discrete analog). This step establishes the complete list of components subject to failure analysis and ensures that no safety-relevant element is omitted from the assessment.

Step 2: Failure Mode Determination. For each component, the possible failure modes are identified based on the component technology and its function within the AMAGI architecture. Each failure mode is classified as one of three categories: safe failure (λ_{safe}), which drives the system toward a safe state; dangerous detectable failure (λ_{dd}), which could cause a dangerous condition but is detected by automatic diagnostics within the proof test interval; and dangerous undetectable failure (λ_{du}), which could cause a dangerous condition and is not detected by any automatic diagnostic. This classification is critical because only λ_{du} contributes to the PFD_{avg} calculation.

Step 3: Failure Rate Estimation. The failure rate for each component is estimated in FIT (1 FIT = 1×10^{-9} failures per hour) using the prediction methodology of IEC 62380:2004. This standard provides base failure rates for electronic component categories (FPGA logic, combinatorial logic, mixed-signal, analog, ROM, OTP, SRAM) adjusted for the specific operating conditions of the AMAGI implementation, including ambient temperature, junction temperature rise, thermal cycling amplitude, and the mission profile of the target application.

Step 4: Diagnostic Coverage Assignment. The diagnostic coverage (DC) for each component is assigned based on the specific diagnostic technique employed, using the reference values in IEC 61508-2 Annex A. The DC represents the fraction of dangerous failures that are detected by the automatic diagnostic measure: $DC = \lambda_{dd} / (\lambda_{dd} + \lambda_{du})$. Conservative values are used throughout; where the Annex A tables provide a range, the lower bound is adopted to avoid overstating the safety performance of the architecture.

Step 5: SFF Computation. The Safe Failure Fraction is computed as SFF = $(\lambda_{safe} + \lambda_{dd}) / \lambda_{total}$, where $\lambda_{total} = \lambda_{safe} + \lambda_{dd} + \lambda_{du}$. The SFF represents the fraction of all failures that either drive the system toward a safe state or are detected by automatic diagnostics. A higher SFF indicates greater resilience against dangerous undetected failures and allows a higher SIL to be claimed for a given hardware fault tolerance.

Step 6: SIL Verification. The computed SFF and the hardware fault tolerance (HFT) of the system architecture are verified against IEC 61508-2 Table 2, which specifies the maximum SIL achievable for a given combination of SFF and HFT. The AMAGI architecture supports two voting configurations: 1oo1 (one-out-of-one, HFT=0), where a single dangerous undetected failure causes loss of the safety function; and 1oo2 (one-out-of-two, HFT=1), where two independent dangerous undetected failures are required to cause loss of the safety function. The 1oo2 architecture is achieved through the EGW AND-gate, which requires both the SCC Enable signal and the external 2FA Enable signal to be present for actuation, effectively providing a dual-channel safety barrier.

4. Component Failure Rate Estimates

This section presents the failure rate estimates for each component of the AMAGI TOE. The failure rates are derived from IEC 62380:2004 prediction methodology, calibrated to the operating conditions of the Xilinx Artix-7 FPGA reference implementation. Each component is characterized by its total failure rate (λ_{total}), the safe failure rate (λ_{safe}), the dangerous detectable failure rate (λ_{dd}), the dangerous undetectable failure rate (λ_{du}), and the diagnostic coverage (DC). The failure mode distribution (safe vs. dangerous) follows the default ratios recommended by IEC 61508-6 for each component type, and the DC values are based on the specific diagnostic techniques implemented in the AMAGI architecture per IEC 61508-2 Annex A.

It is important to note that these are conservative estimates. The IEC 62380 prediction methodology tends to overestimate failure rates for modern FPGA technologies compared to field return data, particularly for devices manufactured on mature process nodes. The Artix-7 family is manufactured on a 28 nm process, which has well-characterized reliability and lower defect density than the assumptions embedded in the IEC 62380 generic models. However, using the standard prediction values provides a defensible, conservative basis for SIL claims.

Component	λ (FIT)	DC (%)	λ_{DU} (FIT)	Method
SCC FSM (core)	60	90	3.0	FMEA+FTA
TTP delay chain	40	80	4.5	FMEA+DDL test
MEM/MAG (BRAM+OTP)	25	99	0.2	ECC+OTP verify
Amagi-Q (PQC core)	30	92	1.8	FMEA+signature
Optocoupled I/O	50	70	8.4	Expert+field

Clock/power isolation	20	85	1.0	FMEA+monitor
Component sum (TCAD baseline)	225		18.9	
EGW (AND-gate) see note	5	99	0.05	BIST truth-table test

Table 2: Component failure rate estimates and diagnostic coverage (per TCAD article Table IV)

Note on the EGW (AND-gate) row: The Executive Gateway (EGW) is a non-programmable hardware AND-gate (Actuator_Power = SCC_Enable AND Ext_2FA_Enable) introduced per ADR-006. Its total failure rate of 5.0 FIT (cited in NIPU v1.1 §3.3) is dominated by input/output buffer and metallization failures on the FPGA implementation; the AND function itself is realised by a single LUT with negligible independent failure modes. The 99% diagnostic coverage is achieved by a power-on BIST that exercises the full AND-gate truth table within 60 seconds and by continuous online monitoring of the SCC_Enable and Ext_2FA_Enable signal integrity. The resulting λ_{DU} = 0.05 FIT reflects only the rare simultaneous failure of both input paths that escapes the BIST detection window. The EGW is listed here for traceability with NIPU v1.1 §3.3 (which cites 5.0 FIT total failure rate) and to make explicit that the EGW is included in the AMAGI FMEDA scope. However, the "Component sum (TCAD baseline)" row above retains the 225 FIT / 18.9 FIT totals from the TCAD reference article Table IV, which did not separately itemise the EGW; the EGW's 5 FIT contribution is therefore treated as a conservative disclosure additional to the baseline rather than as a replacement of it. The SFF computation in §6 uses the TCAD baseline (225 FIT / 18.9 FIT) so that the resulting SFF = 91.6% (component) / 93% (system) remains consistent with the values reported in NIPU v1.1, Framework v2.0 §17, OpenSpec v1.1, HARA v1.0 and the NIST AI RMF Profile v1.1. Including the EGW row explicitly would change SFF to 91.8% (component), which does not alter the SIL classification (still ≥90%, SIL 2 with HFT = 0) but would create a documentation inconsistency across the AMAGI suite.

Note on the MEM/MAG (BRAM+OTP) diagnostic coverage: The MEM/MAG subsystem achieves the highest DC (99%) of any subsystem because ECC single-error correction detects all single-bit and double-bit errors in protected BRAM, while OTP read-back verification at boot combined with CRTM Golden Hash comparison detects nearly all dangerous OTP failures prior to operational use. The resulting λ_{DU} = 0.2 FIT is the lowest of any subsystem, reflecting the comprehensive detection capability of the combined ECC and OTP verification mechanisms.

5. Diagnostic Coverage Analysis

This section provides a detailed analysis of the diagnostic mechanisms employed for each AMAGI subsystem grouping, justifying the DC values assigned in Table 2. The diagnostic coverage values are derived from IEC 61508-2 Annex A, which provides reference DC values for specific diagnostic techniques. Where multiple diagnostic techniques are applied to a single subsystem, the combined DC is computed conservatively (i.e., the highest single-technique DC is used rather than the theoretical combined coverage, unless the techniques are provably independent). All diagnostic measures are implemented in hardware or firmware within the AMAGI Secure World, ensuring that they cannot be bypassed or disabled by the Normal World cognitive clusters.

5.1 SCC FSM (core) (DC = 90%)

The Safety Control Core FSM is the central sequencing element of the AMAGI architecture, controlling the system state transitions among INIT, IDLE, ARMED, ACTIVE, FAULT, and CUTOFF states. The primary diagnostic mechanism is FMEA (Failure Mode and Effects Analysis) combined with FTA (Fault Tree Analysis), which systematically identifies all failure modes of the SCC FSM and traces their effects through the system. Online FSM state transition monitoring verifies that every state transition follows a legal path as defined by the SCC state transition table; any illegal transition triggers an immediate transition to the FAULT state. Additionally, SPIN model checking of the SCC FSM (25,927,680 reachable states, 15 LTL properties (9 safety + 2 architectural assumptions + 4 liveness)) provides formal verification that the FSM design is free from deadlock, livelock, and assertion violations for the 9 safety properties (zero violations); the 4 liveness properties (SCC-L01 through SCC-L03) produce counterexamples without fairness assumptions (FA1, FA2), and SCC-L04 remains open under the current model. The DC of 90% is assigned per IEC 61508-2 Annex A for "FMEA+FTA" combined with "online monitoring of program sequence" and "formal verification." The remaining 10% of dangerous failures are attributed to subtle timing violations that may not cause an illegal state transition but could result in a transient incorrect output.

5.2 TTP Delay Chain (DC = 80%)

The TTP delay chain comprises the PVT-calibrated Digital Delay Lock and the Actuation Time Lock, which together enforce the Tautological Truth Predicate by requiring that a configurable delay elapses before the actuation command is forwarded. The primary diagnostic mechanism is FMEA combined with a DDL test: the PVT calibration with replica-path feedback tracks the current PVT corner and adjusts the DDL delay accordingly, while the delay bound checking mechanism verifies that the DDL output delay remains within the specified tolerance window during each calibration cycle. Additionally, SPIN model checking of the ATL state machine formally verifies that the ATL cannot be bypassed, shortened, or circumvented. The DC of 80% is assigned per IEC 61508-2 Annex A for "FMEA+DDL test" of a mixed-signal component, which provides lower coverage than purely digital diagnostics because analog drift may escape detection between calibration points. The remaining 20% accounts for PVT transients faster than the calibration cycle and for common-mode drift affecting both the reference chain and the operational chain simultaneously. The TTP delay chain contributes λ_{DU} = 4.5 FIT, the second-highest dangerous undetected failure rate among the six subsystems.

5.3 MEM/MAG (BRAM+OTP) (DC = 99%)

The MEM/MAG subsystem combines the Secure Hardware Memory Firewall with ECC protection, the Mutable Authenticated Graph OTP array, and the Code Root of Trust Mask ROM into a single subsystem grouping. This subsystem achieves the highest diagnostic coverage (99%) of any AMAGI subsystem through three complementary mechanisms: (1) ECC single-error correction, which detects and corrects all single-bit errors and detects all double-bit errors in protected BRAM; (2) OTP read-back verification at boot, which confirms that the OTP programming is consistent with the expected values; and (3) CRTM Golden Hash comparison, where the CRTM computes a SHA-256 hash over the MAG OTP content and compares it against the embedded Golden Hash. The combination of these mechanisms provides near-complete detection of dangerous memory failures, because any bit-level corruption will cause either an ECC detection, a read-back mismatch, or a hash mismatch. The DC of 99% is assigned per IEC 61508-2 Annex A for "ECC+OTP verify" of a memory subsystem. The resulting λ_{DU} = 0.2 FIT is the lowest of any subsystem.

5.4 Amagi-Q — PQC Core (DC = 92%)

The Amagi-Q cryptographic engine implements the post-quantum cryptographic suite (FIPS 203 ML-KEM, FIPS 204 ML-DSA, FIPS 205 SLH-DSA) used for audit trail signing, key establishment, and attestation. The primary diagnostic mechanism is FMEA combined with a signature self-test: a known-answer test vector suite exercises the cryptographic algorithms with pre-computed input/output pairs, verifying that the Amagi-Q hardware produces the correct results for each test case. Additionally, a signature verification self-test signs a known message and verifies the signature, confirming end-to-end correct operation of the signing and verification paths. The DC of 92% is assigned per IEC 61508-2 Annex A for "FMEA+signature" self-verification of a programmable logic component. The 8% residual accounts for subtle computational errors that may not be exercised by the test vector suite, such as side-channel leakage that could compromise the cryptographic keys without affecting the functional output.

5.5 Optocoupled I/O (DC = 70%)

The optocoupled I/O subsystem comprises the Emergency Gate Watchdog AND-gate, the unidirectional audit diode, and all galvanically-isolated input/output interfaces. This subsystem has the highest total failure rate (λ = 50 FIT) and the lowest diagnostic coverage (DC = 70%) of any AMAGI subsystem, making it the dominant contributor to dangerous undetected failures (λ_{DU} = 8.4 FIT, representing 44% of the total λ_{DU}). The diagnostic mechanism is expert judgment combined with field data: expert analysis of the optocoupler failure modes (LED degradation, phototransistor degradation, isolation breakdown) supplemented by field return data from similar optocoupler applications in safety-critical systems. The DC of 70% is assigned per IEC 61508-2 Annex A for "Expert+field" data of a galvanically-isolated I/O subsystem. The 30% residual reflects the inherent difficulty of detecting gradual optocoupler degradation (especially LED CTR degradation) and the limited ability to test the isolation barrier during normal operation. This subsystem represents the primary target for future diagnostic coverage improvement.

5.6 Clock/Power Isolation (DC = 85%)

The clock/power isolation subsystem comprises the Analog Fallback Circuit, independent LDO, and clock distribution network. The diagnostic mechanism is FMEA combined with hardware monitoring: the AFC provides watchdog heartbeat monitoring that activates the hardware bypass when the SCC heartbeat signal is absent for longer than the timeout period, while the independent LDO is monitored for output voltage within specification, and the clock distribution network is monitored for frequency and jitter within tolerance. The AFC operates from a separate power rail with an independent LDO, ensuring that it remains functional even if the main digital power domain fails. The DC of 85% is assigned per IEC 61508-2 Annex A for "FMEA+monitor" of a power/clock isolation subsystem. The 15% residual accounts for common-cause failures affecting both the primary power domain and the AFC independent LDO simultaneously, and for clock distribution failures that may not be detected by the frequency monitor within the monitoring interval.

Subsystem	λ (FIT)	DC (%)	λ_{DU} (FIT)	Method
SCC FSM (core)	60	90	3.0	FMEA+FTA
TTP delay chain	40	80	4.5	FMEA+DDL test
MEM/MAG (BRAM+OTP)	25	99	0.2	ECC+OTP verify
Amagi-Q (PQC core)	30	92	1.8	FMEA+signature
Optocoupled I/O	50	70	8.4	Expert+field
Clock/power isolation	20	85	1.0	FMEA+monitor
Sum	225		18.9	

Table 3: Diagnostic coverage summary per TCAD article Table IV

6. SFF Calculation

The Safe Failure Fraction (SFF) is the key metric used by IEC 61508 to determine the maximum SIL achievable by a hardware subsystem for a given hardware fault tolerance. The SFF represents the fraction of all failures that either drive the system toward a safe state or are detected by automatic diagnostics, and is defined as SFF = $(\lambda_{safe} + \lambda_{dd}) / \lambda_{total} = 1 - \lambda_{DU} / \lambda$. A higher SFF indicates that a larger proportion of dangerous failures are detected, reducing the probability that a dangerous undetected failure will accumulate and cause loss of the safety function. This section computes the component-level and system-level SFF for the AMAGI TOE.

6.1 Component-Level SFF

The component-level aggregate failure rates are obtained by summing the corresponding failure rates across all six subsystem groupings of the AMAGI TOE:

Parameter	Summation	Result (FIT)
λ_{total}	60 + 40 + 25 + 30 + 50 + 20	225
λ_{DU}	3.0 + 4.5 + 0.2 + 1.8 + 8.4 + 1.0	18.9

Table 4: Component-level aggregate failure rate summation

6.2 Component-Level SFF Computation

Applying the SFF formula to the component-level aggregate failure rates:

$$SFF_{comp} = 1 - \lambda_{DU} / \lambda_{total} = 1 - 18.9 / 225 = 1 - 0.084 = 91.6\%$$

6.3 System-Level SFF with CRTM/EGW Uplift

The component-level SFF of 91.6% does not account for the additional diagnostic coverage provided by the CRTM (Code Root of Trust Mask ROM) and EGW (Emergency Gate Watchdog) architectural mechanisms, which operate at the system level rather than at the individual component level. Per ADR-006 and ADR-007, the CRTM boot verification and EGW dual-channel actuation gate provide system-level diagnostic uplift that reduces the effective system-level λ_{DU} from 18.9 FIT to approximately 15.75 FIT. This reduction accounts for the fact that the CRTM boot verification detects additional dangerous failures in the MEM/MAG and Amagi-Q subsystems during the boot-time proof test, and the EGW AND-gate provides an independent safety barrier that detects certain dangerous I/O failures before they can cause hazardous actuation.

The system-level SFF is computed as:

$$SFF_{sys} = 1 - \lambda_{DU,sys} / \lambda_{total} = 1 - 15.75 / 225 = 1 - 0.070 = 93\%$$

The system-level SFF of 93% is used throughout the remainder of this document as the conservative SFF figure. Per IEC 61508-2 Table 2, the architectural constraints for Type B components (complex components with software-interacting behavior, which includes all AMAGI FPGA-based subsystems) are as follows: SFF ≥ 90% with HFT = 0 (1oo1 architecture) allows SIL 2; SFF ≥ 90% with HFT = 1 (1oo2 architecture) allows SIL 3. Therefore, the AMAGI architecture with system-level SFF = 93% meets the SIL 2 architectural constraint under the 1oo1 configuration and the SIL 3 architectural constraint under the 1oo2 voting configuration.

7. PFD_{avg} Calculation

The average Probability of Dangerous Failure on Demand (PFD_{avg}) quantifies the likelihood that the safety function will fail to perform its intended action when a demand arises. The PFD_{avg} depends on the dangerous undetected failure rate (λ_{DU}), the proof test interval (TI), and the voting architecture. IEC 61508-1 Table 3 defines the SIL thresholds for low-demand mode in terms of PFD_{avg}: SIL 2 requires PFD_{avg} in the range 10^{-3} to 10^{-4}; SIL 3 requires PFD_{avg} in the range 10^{-4} to 10^{-5}. This section computes the PFD_{avg} for both the 1oo1 and 1oo2 voting architectures, using the system-level $\lambda_{DU,sys}$ = 15.75 FIT that incorporates the CRTM/EGW diagnostic uplift per ADR-006/007.

7.1 1oo1 Architecture (HFT = 0)

For a 1oo1 (one-out-of-one) architecture with a single channel, the PFD_{avg} is computed using the simplified formula from IEC 61508-6:

$$PFD_{avg}(1oo1) = \lambda_{DU,sys} \times T_1 / 2$$

where T_1 is the proof test interval, defined as the interval between complete tests that can detect all dangerous undetected failures. For the AMAGI architecture, the proof test interval is set to 1 year (8,760 hours), corresponding to an annual maintenance cycle during which the CRTM boot verification, MAG OTP read-back, and Amagi-Q test vectors are all exercised. Substituting the values:

$$PFD_{avg}(1oo1) = 15.75\times10^{-9} \times 8{,}760 / 2 = 6.9\times10^{-5}$$

$$PFD_{avg}(1oo1) = 6.9\times10^{-5} \text{ (SIL 2 per IEC 61508-3 Clause 7.3, range } 10^{-3} \text{ to } 10^{-4}\text{)}$$

The 1oo1 architecture achieves SIL 2 based on the PFD_{avg} calculation. The architectural constraint (SFF = 93% ≥ 90%, HFT = 0) also limits the 1oo1 architecture to SIL 2 per IEC 61508-2 Table 2, so both the PFD constraint and the architectural constraint are consistent for the 1oo1 case.

7.2 1oo2 Architecture (HFT = 1)

For a 1oo2 (one-out-of-two) architecture with hardware fault tolerance HFT = 1, two independent channels must fail simultaneously for the safety function to be lost. In the CCF-dominated regime, the PFD_{avg} for the 1oo2 architecture is computed using the common-cause failure model:

$$PFD_{avg}(1oo2) = \beta \times \lambda_{DU,sys} \times T_1 / 2$$

where β is the common-cause failure (CCF) factor, representing the fraction of dangerous undetected failures that can defeat both channels simultaneously. A conservative value of β = 0.1 is used, consistent with IEC 61508-6 Annex D for systems with moderate CCF defenses (separate clock domains via DTI, separate PLL, separate LDO). Substituting the values:

$$PFD_{avg}(1oo2) = 0.1 \times 15.75\times10^{-9} \times 8{,}760 / 2 \approx 6.9\times10^{-6}$$

The 1oo2 architecture achieves $PFD_{avg} = 6.9\times10^{-6}$, which places the AMAGI architecture firmly in the deep SIL 3 range (10^{-5} to 10^{-6}), providing a margin of approximately one order of magnitude above the SIL 3 lower bound. This margin is significant because it accommodates uncertainties in the failure rate predictions and provides a robust basis for the SIL 3 claim even under pessimistic assumptions.

Architecture	HFT	Formula	PFD_{avg}	SIL
1oo1	0	$\lambda_{DU,sys} \times T_1 / 2$	6.9×10^{-5}	SIL 2
1oo2	1	$\beta \times \lambda_{DU,sys} \times T_1 / 2$	6.9×10^{-6}	SIL 3

Table 5: PFD_{avg} results for 1oo1 and 1oo2 architectures

> Note on proof test interval: The proof test interval of TI = 8,760 hours (1 year) assumes an annual maintenance cycle during which the complete proof test (CRTM boot verification, MAG OTP read-back, Amagi-Q test vectors, EGW truth table test) is performed. If a shorter proof test interval is achievable (e.g., through automated online proof testing), the PFD_{avg} can be further reduced proportionally. A proof test interval of 6 months (4,380 hours) would reduce the 1oo1 PFD_{avg} by approximately 50%.

8. Hardware Architectural Constraints Verification

IEC 61508-2 Table 2 specifies the maximum Safety Integrity Level that can be claimed for a hardware subsystem based on its Safe Failure Fraction and hardware fault tolerance. This table defines the architectural constraints that must be satisfied independently of the PFD_{avg} calculation — both the architectural constraints and the PFD constraint must be met for a SIL claim to be valid. The AMAGI hardware components are classified as Type B per IEC 61508-2 §7.4.4.3, because they are complex components with software-interacting behavior (the SCC FSM, MPE, and ATL have state machines that interact with firmware and external inputs, and the Amagi-Q crypto engine implements algorithmic functions with test vectors).

For Type B components, IEC 61508-2 Table 2 imposes the following architectural constraints on the maximum achievable SIL based on SFF and HFT. These constraints are more restrictive than for Type A components (simple, well-defined failure modes) because Type B components have failure modes that are more difficult to predict and model. The complete constraint table for Type B components is reproduced below with the AMAGI operating points highlighted.

SFF	HFT = 0 (1oo1)	HFT = 1 (1oo2)	HFT = 2 (1oo3)
SFF < 60%	Not allowed	SIL 1	SIL 2
60% ≤ SFF < 90%	SIL 1	SIL 2	SIL 3
90% ≤ SFF < 99% — AMAGI (93%)	SIL 2	SIL 3	SIL 4
SFF ≥ 99%	SIL 3	SIL 4	SIL 4

Table 6: IEC 61508-2 Table 2 — Architectural constraints for Type B components (AMAGI operating point highlighted)

8.1 1oo1 Verification (HFT = 0)

With the 1oo1 architecture (HFT = 0), the AMAGI hardware achieves SFF = 93% ≥ 90%, which allows a maximum SIL of SIL 2 per IEC 61508-2 Table 2. The $PFD_{avg} = 6.9\times10^{-5}$ also satisfies the SIL 2 PFD range (10^{-3} to 10^{-4}). Both the architectural constraint and the PFD constraint are consistent, confirming the SIL 2 claim for the 1oo1 architecture. To achieve SIL 3, the hardware fault tolerance must be increased to HFT = 1 through the dual-channel voting architecture provided by the EGW AND-gate.

8.2 1oo2 Verification (HFT = 1)

With the 1oo2 architecture (HFT = 1) provided by the EGW AND-gate, the AMAGI hardware achieves SFF = 93% ≥ 90% with HFT = 1, which allows a maximum SIL of SIL 3 per IEC 61508-2 Table 2. The $PFD_{avg} = 6.9\times10^{-6}$ also satisfies the SIL 3 PFD range. Both the architectural constraint and the PFD constraint are satisfied, confirming the SIL 3 claim for the 1oo2 architecture.

8.3 Route 2_H Compliance

IEC 61508-2 provides two routes for demonstrating compliance with the hardware safety integrity requirements. Route 1_H is based solely on the SFF and HFT values from Table 2. Route 2_H allows a higher SIL to be claimed if the component has a proven-in-use record demonstrating a sufficiently low dangerous failure rate based on field return data. The AMAGI architecture claims compliance via Route 2_H based on the following argument: SFF ≥ 90% with HFT = 1 → SIL 3, supported by the conservative failure rate predictions of IEC 62380 and the formal verification evidence (SPIN model checking) that reduces the probability of systematic design errors in the safety-critical state machines. The combination of conservative failure rate prediction and formal verification provides a strong basis for the Route 2_H claim, even without extensive field return data for the AMAGI-specific implementation.

9. Conclusions

This FMEDA demonstrates that the AMAGI hardware architecture achieves a level of safety integrity that meets the requirements for SIL 2 under the 1oo1 voting architecture and SIL 3 under the 1oo2 voting architecture, as defined by IEC 61508-2. The analysis has been conducted systematically in accordance with IEC 61508-2 §7.4.6 and §7.4.9, using failure rate predictions per IEC 62380:2004 and diagnostic coverage values derived from IEC 61508-2 Annex A. The results provide a comprehensive, quantitative basis for the AMAGI safety argument and support the certification submission for functional safety compliance.

The key results of the FMEDA are summarized in the following table. These results confirm that the AMAGI hardware architecture satisfies both the architectural constraints and the PFD constraints for SIL 2 (1oo1) and SIL 3 (1oo2). The system-level SFF of 93% incorporates the CRTM/EGW diagnostic uplift per ADR-006/007, which reduces the effective system-level λ_{DU} from 18.9 FIT (component-level) to ~15.75 FIT (system-level).

Metric	Value	Requirement	Compliance
SFF (component-level)	91.6%	≥ 90% for Route 2_H	PASS
SFF (system-level)	93%	≥ 90% for Route 2_H + HFT=1	PASS
PFD_{avg} (1oo1)	6.9×10^{-5}	10^{-3} to 10^{-4} (SIL 2)	PASS
PFD_{avg} (1oo2)	6.9×10^{-6}	10^{-4} to 10^{-5} (SIL 3)	PASS
Minimum DC	70%	≥ 60% per component	PASS
Maximum DC	99% (MEM/MAG)	N/A	PASS
λ_{DU} (component-level)	18.9 FIT	N/A (lower is better)	PASS
λ_{DU} (system-level)	~15.75 FIT	N/A (lower is better)	PASS

Table 7: FMEDA results summary — SIL 2 (1oo1) / SIL 3 (1oo2) compliance verification

The analysis further reveals the following key findings regarding the AMAGI hardware architecture. All six subsystem groupings have diagnostic coverage ≥ 70%, with the highest DC (99%) achieved by the MEM/MAG (BRAM+OTP) subsystem through the combination of ECC protection, OTP read-back verification, and CRTM Golden Hash comparison. The optocoupled I/O subsystem has the highest total failure rate (λ = 50 FIT) and the lowest DC (70%), making it the dominant contributor to λ_{DU} (8.4 FIT, 44% of component-level λ_{DU}). The TTP delay chain is the second-largest contributor (λ_{DU} = 4.5 FIT, 24% of component-level λ_{DU}). Together, the optocoupled I/O and TTP delay chain account for approximately 68% of all component-level dangerous undetected failures. These two subsystems represent the primary targets for future diagnostic coverage improvement, which could be achieved through additional online monitoring techniques, redundant I/O channels, or improved DDL calibration coverage.

The AMAGI hardware architecture achieves SIL 3 compliance under the 1oo2 voting architecture with a margin of approximately one order of magnitude above the SIL 3 lower bound (PFD_{avg} = 6.9×10^{-6} vs. the SIL 3 lower bound of 10^{-5}). Under the 1oo1 architecture, the system achieves SIL 2 with PFD_{avg} = 6.9×10^{-5}. The dual-channel EGW AND-gate provides the HFT = 1 necessary for the SIL 3 claim. This margin provides confidence that the SIL 3 claim is robust against moderate increases in failure rates due to environmental stress, aging, or manufacturing variability. The results confirm that the AMAGI Architectural Class hardware is suitable for deployment in safety-critical applications requiring SIL 3 functional safety integrity, including medical implant controllers (IEC 60601-1), industrial safety systems (IEC 61508), and automotive safety elements (ISO 26262 ASIL D equivalent).

Recommendations for future work: (1) Reduce the optocoupled I/O λ_{DU} contribution through redundant I/O channels or improved optocoupler monitoring to further increase the SFF margin; (2) Improve the TTP delay chain diagnostic coverage from 80% to ≥ 90% through continuous DDL calibration or dual-delay-chain comparison; (3) Conduct field return data collection for the Artix-7 FPGA implementation to support the Route 2_H proven-in-use argument with empirical evidence; (4) Investigate automated online proof testing to reduce the effective proof test interval from 8,760 hours to a shorter interval, further reducing PFD_{avg}; (5) Extend the FMEDA to include common cause failure (CCF) analysis per IEC 61508-6 Annex D, quantifying the beta factor for the 1oo2 architecture with explicit consideration of the DTI, separate PLL, and separate LDO measures.